

infer the performance of advertising

campaigns, conversion rates, etc

- IoT and device metrics
- Druid can be effectively used for business intelligence. With the ability to support high concurrency it becomes an ideal tool for interactive visual analytics.

The above is merely a sample list.

Druid can be used in any scenario with identical criteria.

Key features of Druid

The inherent design of Druid makes it suitable for many scenarios where highly reliable, concurrent processing of massive real-time data is required. A few key features of Druid are listed below.

- **Column based storage:** Druid loads only those columns that are needed in the current scenario. The columns are also optimised for storage as per their data types. This enables quick scans and aggregation operations.
- **Scalable distributed system:** An important feature of Druid is its ability to be deployed in a distributed manner across tens or hundreds of servers.
- **Massive parallel processing:** Druid can process a given query in parallel mode across the entire cluster. This helps it to reduce the latency as much as possible.
- **Support for various types of data ingestion:** Druid supports both real-time ingestion (explained later) and batch ingestion of data. In the case of real-time ingestion, the data becomes immediately available for querying.
- **Self-healing, self-balancing and easy to operate:** Druid's inherent design is such that if any of its servers fail then the request can be routed around to provide uninterrupted service 24x7. The official documentation states that Druid is designed to avoid planned downtimes for performing routines

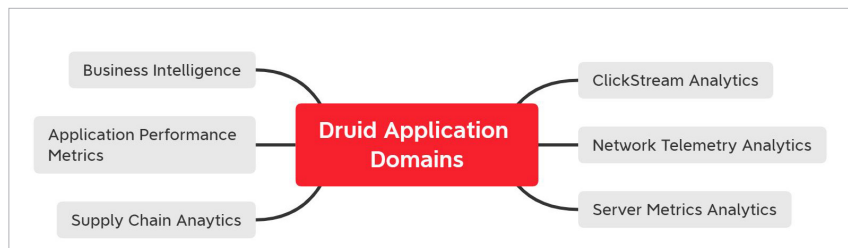


Figure 2: Application domains for Druid

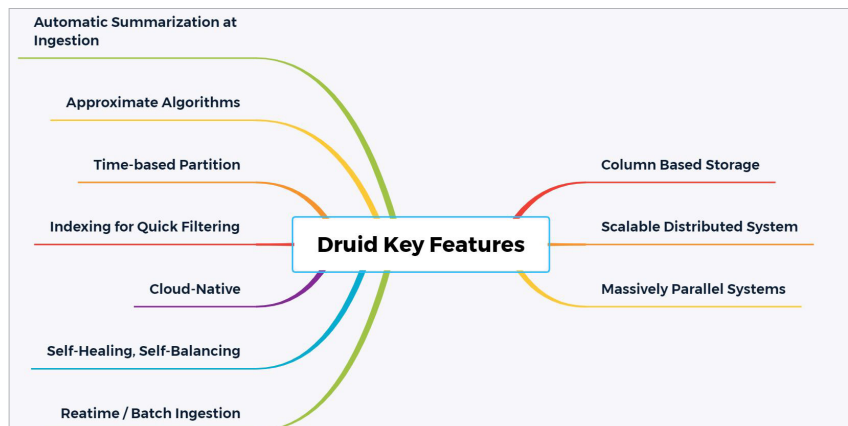


Figure 3: Key features of Druid

such as software updates and configuration enhancement.

- **Cloud native and fault-tolerant:** After the ingestion of data, it becomes fault-tolerant and fail-safe by storing a copy in deep storage. (This is usually HDFS, cloud storage or a shared file system.) The data stored in this deep storage can be utilised in case Druid servers fail. The replication of data storage makes the recovery of systems simple and effective.
- **Indexing for quicker retrieval:** Druid employs indexing mechanisms such as 'roaring'. These enable powerful and quick filtering of data across multiple columns.
- **Time based partitioning:** Druid has the ability to partition data as per time first, and then based on fields. This enables efficient handling of time series data by accessing those specific partitions that match the temporal range of the current query.

Approximate algorithms:

Another important attribute of Druid is the inclusion of algorithms to perform approximate distinct count and approximate ranking, as well as to build approximate histograms and quantiles.

Automatic summarisation during ingestion:

Druid has the ability to perform data summarisation at ingestion time. Because of this, pre-aggregation is carried out to a certain extent. This plays a significant role in increasing the performance of Druid.

Powered by Druid

The power of Druid can be understood by looking at the long list of industry majors who use it. The complete list is available on the official website at <https://druid.apache.org/druid-powered>. Due to space constraints this article lists only a few of the major implementations.